

NANDHINI K

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PROFESSIONAL SUMMARY

Motivated B.E. CSE student (CGPA: 8.1) with hands-on experience in AI-driven IoT systems and practical knowledge of Python programming. Actively building proficiency in Data Structures and Algorithms (DSA) and strengthening foundational skills in Artificial Intelligence and Machine Learning. Built a real-world AI-powered hardware+software project using ESP32 and sensor fusion, combining embedded systems with intelligent software design.

TECHNICAL SKILLS

Languages: Python, SQL, C programming

Skills: Frontend, Ionic Framework (React)

Design & Tools: Figma, UI/UX Design, Wireframing, Prototyping, Usability Testing

INTERNSHIP EXPERIENCE

UI/UX Design Intern | EduTant (2024)

- Completed a hands-on internship focused on building practical UI/UX design skills using Figma
- Learned wireframing, prototyping, and interactive design fundamentals through practical, tool-based training
- Gained hands-on insight into UI flow design, layout principles, and how design decisions impact usability and navigation

PROJECTS

AI-Powered Seasonal Zone-Based Energy Optimization System (ESP32 | Python | IoT | AI | 2026)

College Innovation Project — Innovators' Day 2K26, AIHT

- Built a **smart energy system** using **ESP32** with **multi-sensor fusion (DHT11, LDR, PIR, ACS712)** for **real-time occupancy and environment sensing**.
- Implemented an **AI Decision Engine** minimizing $J = \alpha \times \text{Energy} + \beta \times \text{Discomfort}$ to autonomously control **PWM fans and LEDs** across **two independent zones**.
- Developed a **live WiFi dashboard** with **voice control** via **Chrome Speech Recognition API**; achieved **zone-level energy savings** by **auto-switching devices in unoccupied zones**.

UI/UX Design using Figma Tools (Figma | User Research | 2024)

- Designed **4 end-to-end mobile app interfaces** — **Shopping, Food Delivery, Hotel Booking, and Beauty Product** — focusing on **intuitive user flows** and **clean visual hierarchy**.
- Created **wireframes, prototypes** using **Figma** with **consistent design systems, reusable components, and responsive layouts**.
- Applied **UX principles** including **user research, persona mapping, and usability best practices** to deliver **user-centered designs**.

EDUCATION

B.E. Computer Science and Engineering — CGPA: 8.16/10

Anand Institute of Higher Technology (AIHT), Kazhipattur | Affiliated to Anna University

Higher Secondary Certificate (PCM) — 84%

M.P.B Govt Girls Hr. Sec School, East Tambaram

CERTIFICATIONS, HACKATHONS & EVENTS

- Prayathon — 8hr Hackathon, PRAYATNA'26, ACT, MIT Anna University (Mar 2026)
- Innovators' Day 2K26 — AI IoT Project Showcase, AIHT (Mar 2026)
- Chakravyuha'26 — Vibe Code Off, GDG On Campus, MIT Anna University (Mar 2026)
- Freedom with AI — AI & ChatGPT Workshop, Certificate of Completion (Jun 2026)
- Python using AI Workshop — Hands-on training in AI-assisted Python programming, debugging, and code optimization, AI For Techies
- MongoDB — NoSQL Database Fundamentals, MongoDB University (2024)
- UI/UX Internship — Completed at EduTant (2024)
- Technovanza — Paper Presentation, Inter-college Technical Event (2024)